House Price Prediction Using Machine Learning Algorithms

Dataset used:kc_house_data

The House Price Prediction project aims to estimate property prices using machine learning algorithms. It uses the KC House Data dataset from King County, USA, which includes various housing features. The project applies multiple regression models to analyze and predict prices accurately. Data preprocessing and feature scaling are performed to improve model performance. Different algorithms are compared using evaluation metrics like MAE, MSE, and R^2 score. The project helps in understanding real estate trends and supports data-driven decision-making.

Project Goal: The goal of this project is to build a machine learning model that can accurately predict house prices using housing features. It aims to compare different regression algorithms to find the best-performing model.
The project also focuses on improving prediction accuracy through data preprocessing and feature scaling.

The primary libraries/modules required for this project are:

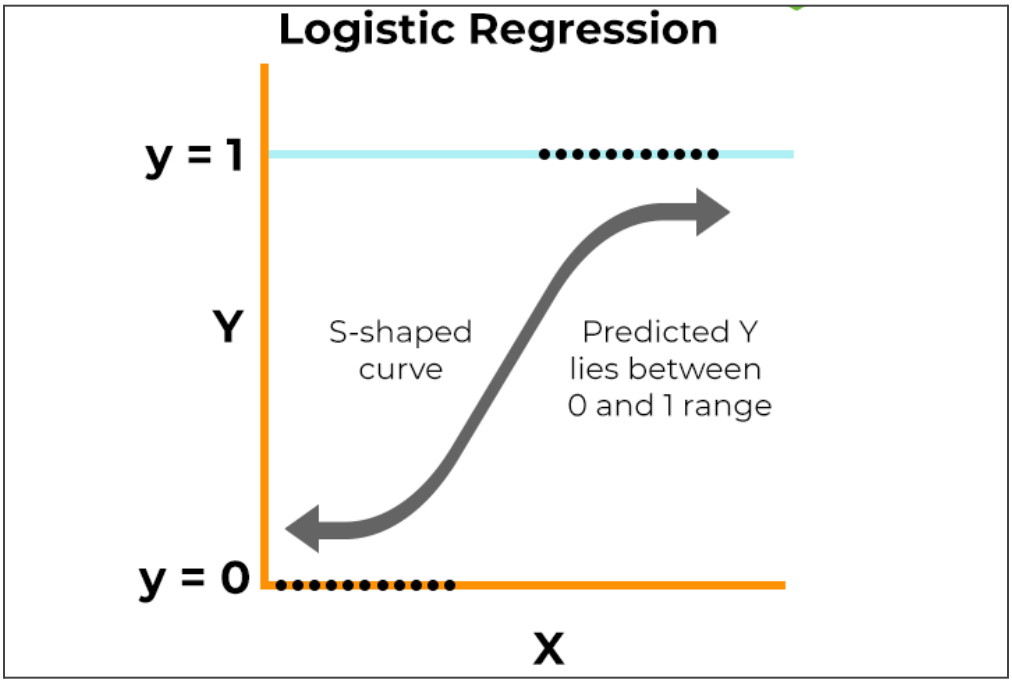
- **NumPy** for numerical operations and array handling.
- **Pandas** for data loading, cleaning, and manipulation.
- **Scikit-learn** for implementing regression algorithms and model evaluation.
- **Matplotlib** for plotting graphs and visualizing data.
- **Seaborn** for advanced and more attractive visualizations.

The Machine Learning Algorithms we Used are:

1. **Logistic Regression**
2. **Naive Bayes (Gaussian NB)**
3. **Decision Tree Regressor**
4. **Random Forest Regressor**
5. **Support Vector Regressor (SVR)**
6. **K-Nearest Neighbors (KNN) Regressor**
7. **Gradient Boosting Regressor**

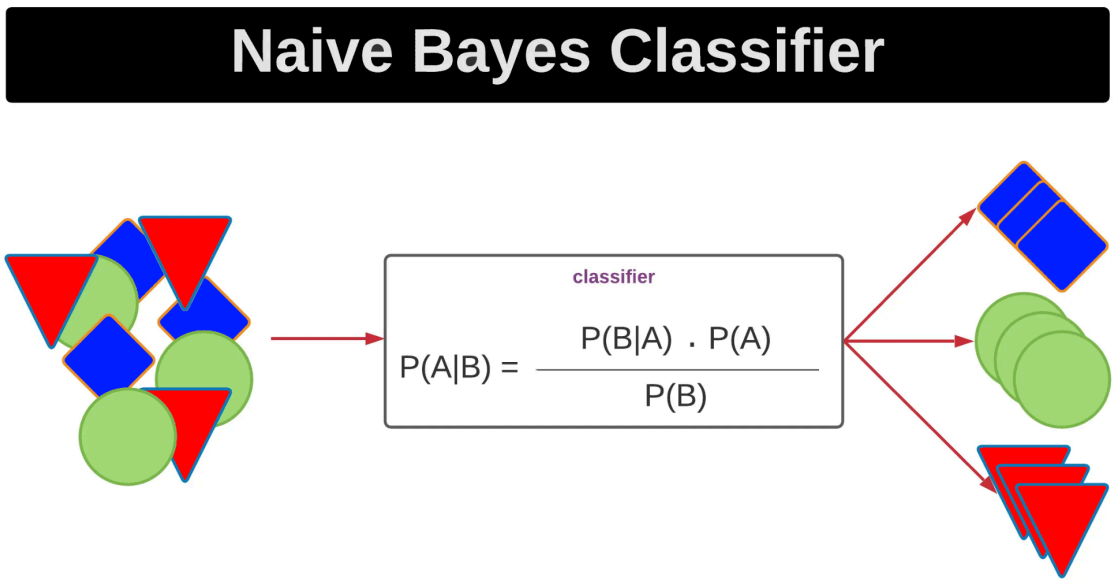
1. **Logistic Regression:**

Logistic Regression converting the continuous house prices into categories such as low, medium, and high. It is then used as a classification algorithm to predict the price category of a house. The model estimates the probability of a house belonging to a particular price class based on features like bedrooms, bathrooms, and square footage. A sigmoid function is used to map predictions between 0 and 1. Based on a threshold, the final class label is assigned. It works best when the relationship between features and target is approximately linear. However, since house price is naturally continuous, regression models are generally more suitable for this dataset.



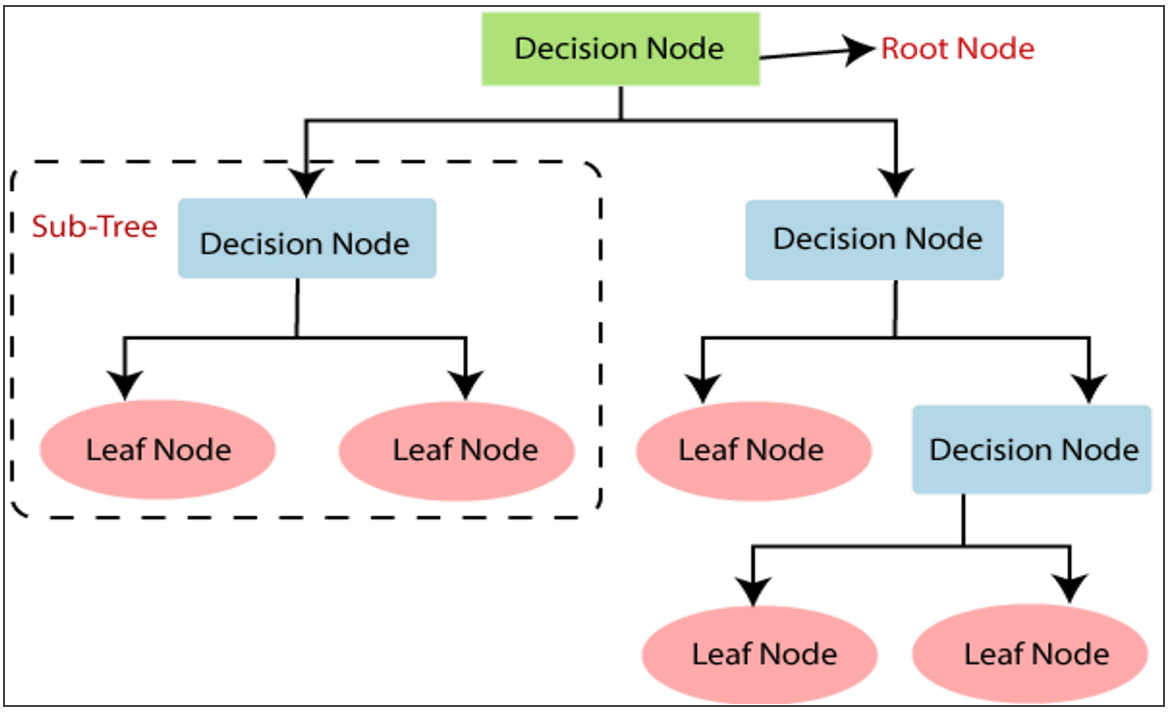
2. **Naive Bayes (Gaussian NB):**

Naive Bayes (Gaussian NB) first converting house prices into categories such as low, medium, and high. It is a classification algorithm based on Bayes' theorem and assumes that all input features are independent. Gaussian NB works well when numerical features like bedrooms, bathrooms, and square footage follow a normal distribution. The model calculates the probability of each price category based on these features. It then predicts the class with the highest probability. It is fast and efficient, especially for large datasets. However, the independence assumption may limit its accuracy for complex house price relationships.



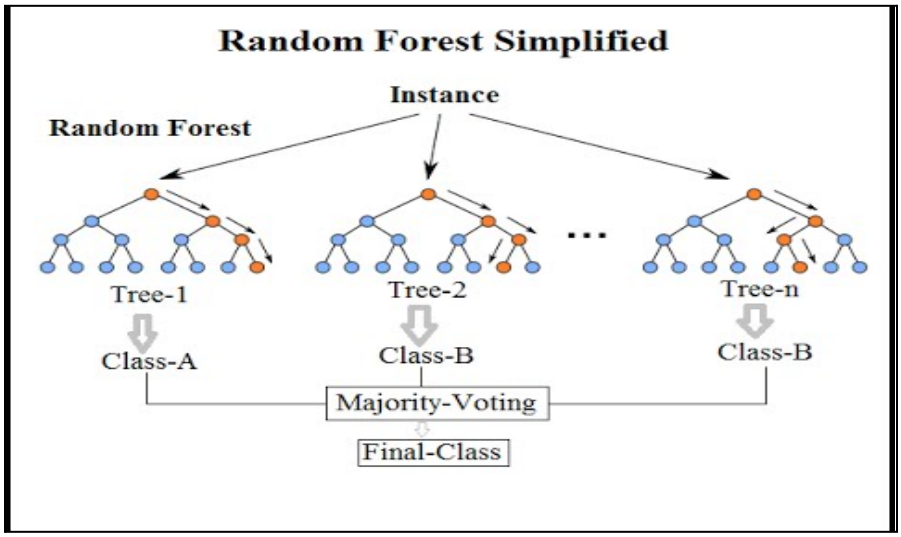
3. **Decision Tree Regressor:**

Decision Tree Regressor is a non-linear algorithm that splits data based on feature values. It builds a tree-like structure where each node represents a decision. The model is easy to understand and visualize. It can capture complex relationships between variables. Decision trees require little data preprocessing. However, they are prone to overfitting if not controlled properly. Techniques like pruning and limiting depth can improve performance. It works well for both small and large datasets. It is widely used due to its simplicity and flexibility.



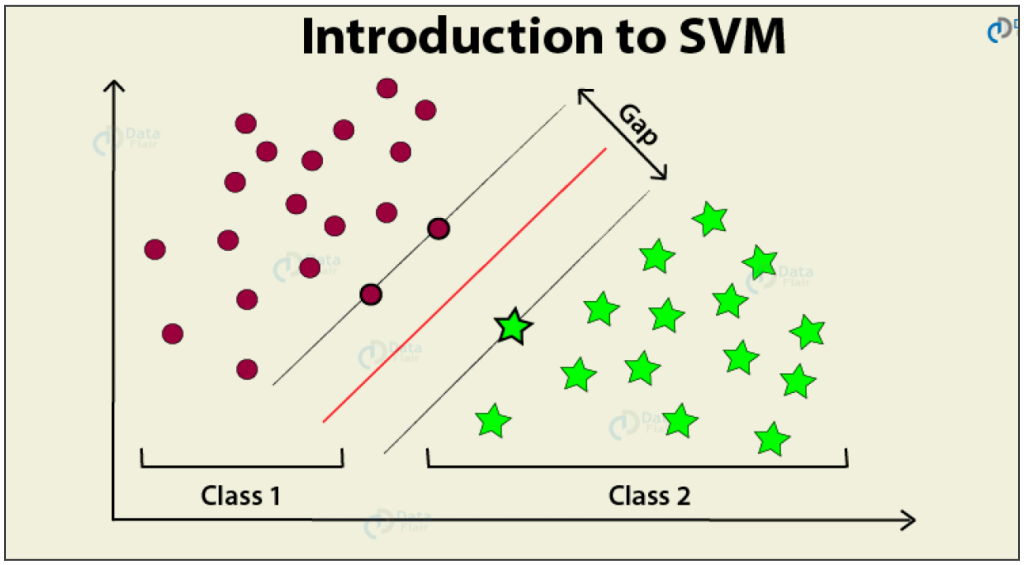
4. **Random Forest Regressor:**

Random Forest Regressor is an ensemble learning method that combines multiple decision trees. Each tree is trained on random subsets of data and features. The final prediction is obtained by averaging outputs of all trees. This reduces overfitting and increases accuracy. It is robust and performs well on large datasets. Random Forest can capture complex and non-linear relationships. It also handles missing values and noise effectively. However, it requires more computational resources. The model is less interpretable compared to a single decision tree.



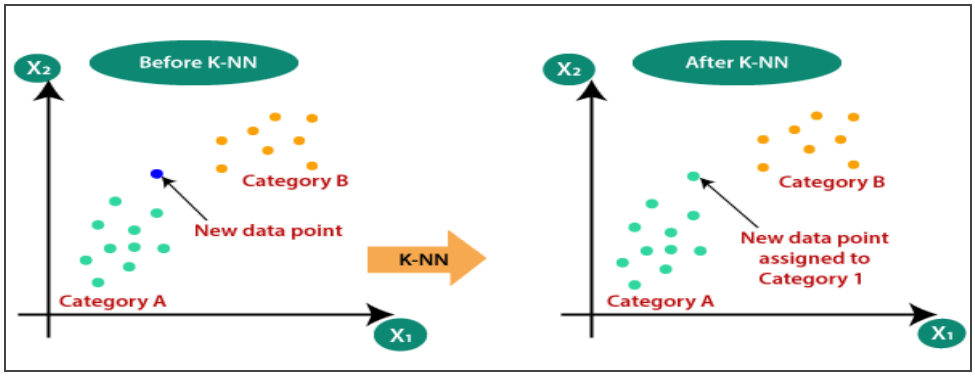
5. **Support Vector Regressor(SVR):**

Support Vector Regressor is based on Support Vector Machine concepts. It tries to find a function that fits data within a margin of tolerance. Instead of minimizing error, it focuses on maintaining a boundary around predictions. SVR works well with high-dimensional data. It can model non-linear relationships using kernel functions. Proper feature scaling is essential for good performance. However, SVR can be slow for large datasets. It is sensitive to parameter tuning. Despite this, it provides strong performance in many regression tasks.



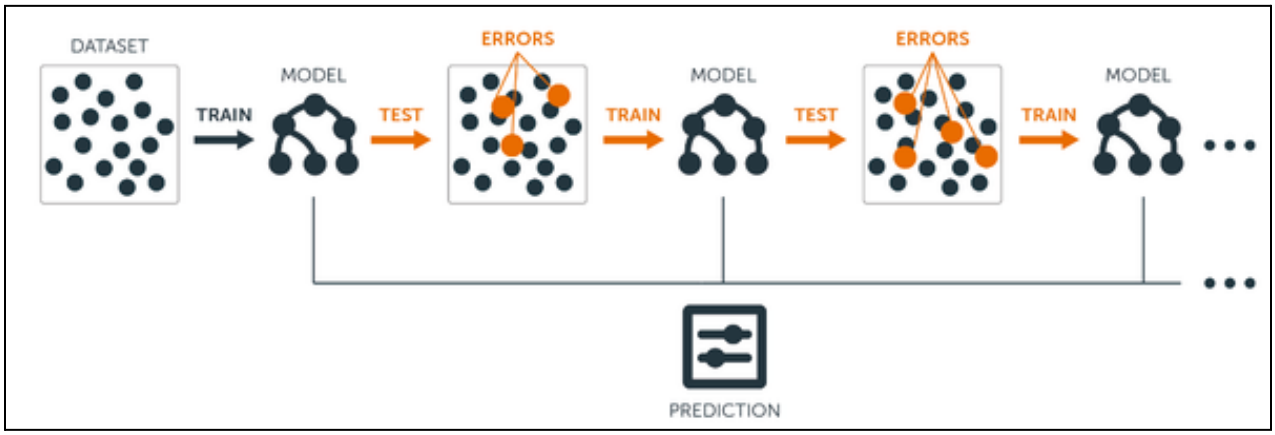
6. **K-Nearest Neighbours (KNN) Regressor:**

KNN Regressor is a simple algorithm that predicts values based on nearby data points. It calculates distances between points to find the nearest neighbors. The prediction is the average of these neighbors. It does not require a training phase, making it easy to use. KNN can capture complex patterns in data. However, it becomes slow as the dataset size increases. It is sensitive to the choice of K value. Feature scaling is important for accurate distance measurement. It works well for smaller datasets.



7. **Gradient Boosting Regressor:**

Gradient Boosting Regressor is an advanced ensemble technique that builds models sequentially. Each new model corrects errors made by previous models. It uses weak learners, usually decision trees, to improve performance. The algorithm focuses on reducing residual errors step by step. It provides high accuracy and strong predictive power. Gradient Boosting handles complex data relationships effectively. However, it requires careful parameter tuning. Training can be time-consuming. It is widely used in real-world machine learning applications.



Conclusion:

The House Price Prediction project successfully applies multiple machine learning algorithms to estimate property prices using the KC House Data dataset. Different models were trained and evaluated using metrics like MAE, MSE, and R^2 score. The comparison of algorithms helps in identifying the most accurate and reliable model. Data preprocessing and feature scaling played an important role in improving performance. Ensemble methods like Random Forest and Gradient Boosting generally provided better results. Overall, the project demonstrates how machine learning can be effectively used for real-world real estate prediction.

What we Learned ?

This project helped in understanding the fundamentals of machine learning using the Iris dataset. I learned how to preprocess data, select features, and split data into training and testing sets. It improved my knowledge of applying different classification algorithms like KNN, Decision Tree, Random Forest, SVM, and Logistic Regression. I also learned how to evaluate model performance using accuracy. The project showed how different algorithms behave on the same dataset. It enhanced my understanding of model comparison and selection. Overall, it strengthened my practical skills in machine learning and Python.